MTH603 Assignment # 02 Solution Fall 2023/2024

Section InCharge: Arooba Fatima

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Solved By: @OneStepToSucess

For Paid MS Word File: wa.me/03191470250

Note: You can follow same procedure to solve your assignment Don't just copy paste check All calculations before submitting, if you found any mistake correct it yourself.

Question #01:

Find $f'(1.40)$ and $f''(1.40)$ by using Newton's Backward difference formula for derivatives from the following table

x	1.24	1.28	1.32	1.36	1.40
$f(x)$	3.2542	4.1562	4.8531	5.4623	6.1811

Solution:

Newton's backward difference table is

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$
1.24	3.2542				
		0.902			
1.28	4.1562		-0.2051		
		0.6969		0.1174	
1.32	4.8531		-0.0877		**0.0799**
		0.6092		**0.1973**	
1.36	5.4623		**0.1096**		
		0.7188			
1.4	**6.1811**				

Given that the differences in x are equal ($\Delta x = 1.36 - 1.32 = 1.32 - 1.28 = 1.28 - 1.24 = 0.04$)

We want to find $f(x)$: $x_n = 1.40$

$$\left[\frac{dy}{dx}\right]_{x=x_n} = \frac{1}{h}\left(\nabla y_n + \frac{1}{2}\nabla^2 y_n + \frac{1}{3}\nabla^3 y_n + \frac{1}{4}\nabla^4 y_n\right)$$

$$\left[\frac{dy}{dx}\right]_{x=1.40} = \frac{1}{0.04}\left(0.7188 + \frac{1}{2}(0.1096) + \frac{1}{3}(0.1973) + \frac{1}{4}(0.0799)\right)$$

$$\left[\frac{dy}{dx}\right]_{x=1.40} = \frac{1}{0.04}(0.7188 + 0.0548 + 0.065766 + 0.01997)$$

$$\left[\frac{dy}{dx}\right]_{x=1.40} = \frac{1}{0.04}(0.859336) = 21.4834$$

$$\left[\frac{d^2y}{dx^2}\right]_{x=x_n} = \frac{1}{h^2}\left(\nabla^2 y_n + \nabla^3 y_n + \frac{11}{12}\nabla^4 y_n\right)$$

$$\left[\frac{d^2y}{dx^2}\right]_{x=1.40} = \frac{1}{0.0016}\left(0.1096 + 0.1973 + \frac{11}{12}(0.0799)\right)$$

$$\left[\frac{d^2y}{dx^2}\right]_{x=1.40} = \frac{1}{0.0016}(0.1096 + 0.1973 + 0.07324)$$

$$\left[\frac{d^2y}{dx^2}\right]_{x=1.40} = \frac{1}{0.0016}(0.38014) = 237.5875$$

Question #02:

Using Simpson's $1/3$ Rule to evaluate

$$\int_0^{12} (e^{-3x} + 0.5)dx$$

(Divide the interval of integration into 8 equal parts.)

Solution:

$$h = \frac{b - a}{N}$$

$$h = \frac{12 - 0}{8} = 1.5$$

The value of table for x and y

x	0	1.5	3	4.5	6	7.5	9	10.5	12
y	0.5	0.5747	0.6494	0.724	0.7987	0.8734	0.9481	1.0228	1.0974

Using Simpsons $\frac{1}{3}$ Rule

$$\int ydx = \frac{h}{3}\left[\left(y_0 + y_8\right) + 4\left(y_1 + y_3 + y_5 + y_7\right) + 2\left(y_2 + y_4 + y_6\right)\right]$$

$$\int ydx = \frac{1.5}{3}[(0.5 + 1.0974) + 4 \times (0.5747 + 0.724 + 0.8734 + 1.0228) + 2 \times (0.6494 + 0.7987 + 0.9481)]$$

$$\int ydx = \frac{1.5}{3}[(0.5 + 1.0974) + 4 \times (3.1949) + 2 \times (2.3962)]$$

$$\int ydx = 9.5847$$

Solution by Simpson's $\frac{1}{3}$ Rule is 9.5847